ASSIGMENT 1.1

1. **Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.**
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2. Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

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Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Speed	Slower (mechanical parts, ~80–160 MB/s)	Much faster (no moving parts, ~200–7000 MB/s)
Durability	Less durable (sensitive to shock & movement)	Highly durable (shock-resistant, no moving parts)
Noise	Noisy (spinning disks & moving heads)	Silent (no mechanical movement)
Power Usage	Consumes more power	Uses less power
Storage Cost	Cheaper per GB	More expensive per GB
Capacity	Typically larger (up to several TB easily)	Usually smaller but growing rapidly
Typical Use Cases	Desktop PCs, bulk storage, backups, servers	Laptops, gaming PCs, high-performance systems

3. **Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.**
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   - **CPU (Central Processing Unit):**
     The CPU acts as the brain of the computer, handling game logic, physics calculations, and player inputs. In games like PUBG or Free Fire, it ensures smooth gameplay by processing actions such as movement, shooting, and interactions in real time.
   - **RAM (Random Access Memory):**
     RAM temporarily stores the data that the game needs while running, such as maps, textures, and active processes. More RAM allows the game to load faster and run smoothly without lag or stuttering, especially during intense gameplay.
   - **GPU (Graphics Processing Unit):**
     The GPU is responsible for rendering images, animations, and visual effects in the game. It ensures high-quality graphics, smooth frame rates, and realistic visuals, making the gaming experience more immersive

4. **Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.**

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Component	**Why Upgrade It?**
CPU (Processor)	Handles game logic, AI, and physics. A faster CPU ensures smoother gameplay, especially during complex match situations.
GPU (Graphics Card)	Renders visuals, players, and stadiums. A better GPU improves frame rates and allows higher graphics settings.
RAM (Memory)	Stores game data while running. More RAM (at least 16GB) reduces lag and improves multitasking.
Storage (SSD)	Speeds up game loading times and system responsiveness compared to HDD.
Cooling System	Prevents overheating during long gaming sessions, ensuring stable performance.
Power Supply (PSU)	Provides sufficient and stable power for upgraded components like GPU and CPU.